Titanium Track Commuter - Size 53 S - Frame Specification

Belt-drive track/commuter frame with the fit and handling geometry of the Schindelhauer Hektor (size 53), re-engineered in titanium. Rev A, 2026-10-21. All dimensions in mm and degrees; files: STEP (3D), DXF 1:1 and PDF (2D).



1. Geometry

Parameter	Hektor 53 (published)	This frame	Note
Stack	498	498.0	kept
Reach	406	406.0	kept
Head tube length	120	120.0	kept; ZS44/ZS56 zero-stack cups
Head tube angle	73.5	73.5	kept
Seat tube angle	75.0	75.0	kept
Seat tube c-t	530	530.0	BB centre to top of seat tube
Top tube (horizontal)	530	539.4	follows from stack/reach/STA; published value is ~9 mm short in every size
Chainstay	378	384-400 (nom. 390)	horizontal slot; 378 only possible with Hektor's shaped aero seat tube
Wheelbase	961	967-983 (nom. 973)	961 at CS 378 -> geometry matches Hektor exactly
Standover	793	781	estimate, 25 mm tyre, at BB
BB drop	n/p	48.4	derived (not published); confirmed from product photo
BB height	n/p	287.6	25 mm tyre
Fork axle-to-crown	n/p	365.0	design value; incl. lower headset
Fork offset	n/p	45.0	design value
Trail	n/p	52.6	25 mm tyre
Front centre	n/p	588.1	toe overlap ~18 mm w/ 170 cranks

Parameter	Hektor 53 (published)	This frame	Note
Headset	1 1/8"	ZS44 / ZS56 tapered	Hektor uses integrated tapered 1 1/8-1.5
Seatpost	27.2	27.2	reamed Ti sleeve in 31.8 seat tube

2. Rear axle position (horizontal track ends)

Chainstay	Wheelbase	Seat-tube / tyre gap (25 mm)	Rear brake reach
384.0	967.0	3.5	46.9
390.0	973.1	9.4	50.5
400.0	983.2	19.1	56.7

Belt centre distance with a 113T belt: 390.3 mm (60/22 freewheel), 392.8 mm (60/21 fixed). Both sit inside the slot.

3. Tube set (proposal - builder to confirm)

Member	Material	Size (OD x wall)	Note
Top tube	3Al-2.5V CWSR	31.8 x 0.9	straight gauge; builder may butt 0.9/0.6/0.9
Down tube	3Al-2.5V CWSR	44.5 x 0.9	oversize for BB/belt stiffness
Seat tube	3Al-2.5V CWSR	31.8 x 0.9	120 mm internal sleeve reamed to 27.2; rear clamp slot + relief hole
Head tube	6Al-4V machined	OD 48/62 tapered	ZS44 top / ZS56 bottom bearing bores (SHIS)
BB shell	6Al-4V	OD 41 x 68 wide	BSA 1.37" x 24 TPI, face & chase
Chainstays	3Al-2.5V CWSR	22.2 x 0.9	crimped to 16 x 27 oval for first 110 mm, then flared
Seat stays	3Al-2.5V CWSR	16.0 x 0.9	bent: straight 85 mm at dropout (belt clearance), then to seat tube
Dropouts	6Al-4V plate	6 mm	horizontal rear-facing track ends, 10.2 slot
Brake bridge	3Al-2.5V	12.7 x 1.0	6 mm hole for recessed-nut caliper
Belt splitter	Ti / builder std.	-	bolted split in drive-side seat stay

4. Interfaces and standards

Item	Specification
Wheels / tyres	700c (622). Designed and checked for 25 mm (stock Ultrasport III 25-622).
Rear hub	120 mm OLD track hub, 3/8" or 10 mm solid axle, horizontal dropouts.
Drivetrain	Gates Carbon Drive CDX, 60T front, 22T (9-spline freewheel) or 21T (track thread fixed), 113T belt, chainline 42.5 mm.
Frame split	Required for belt: bolted splitter in drive-side seat stay (or split dropout).
Bottom bracket	BSA / English threaded 68 mm, external cups (stock: Hollow-Axle 170 mm cranks).
Headset / fork	ZS44/ZS56 tapered head tube; any 1 1/8"-1.5" tapered carbon fork, 365 mm A-C, 45 mm offset, drilled for a rim brake.
Seatpost / clamp	27.2 mm post; 31.8 mm external seat collar.
Brakes	Rear rim-brake bridge for long-reach (47-57 mm) caliper; front through fork crown. Fixed-gear build can omit both (bridge is harmless).
Cable routing	Rear brake internal through top tube: 6 mm ports 45 mm from each end, non-drive side.
Mounts	2 bottle bosses on down tube, 64 mm spacing, lower at 160 mm from BB centre.

5. Manufacturing notes

- TIG weld with full back-purge (argon); no discoloured welds.

- Post-weld: face & chase BB, face/ream head tube bores to ZS44/ZS56, ream seat tube to 27.2.

- Rear dropouts must be parallel and coplanar (belt drive is intolerant of misalignment): check rear triangle alignment to +/-0.5 mm and dropout faces parallel within 0.2 mm.

- Belt line: rear cog and front ring must align within 0.5 mm at 42.5 mm chainline - confirm with the actual hub/crank before welding the chainstays.

- Finish: brushed or bead-blasted raw titanium (no paint). No branding of Schindelhauer.

6. Open items to agree with the builder

- Builder to confirm tube gauges/butting and the chainstay crimp (or substitute a machined yoke).

- 28 mm tyres would need a wider chainline or a yoke; the current layout is checked for 25 mm.

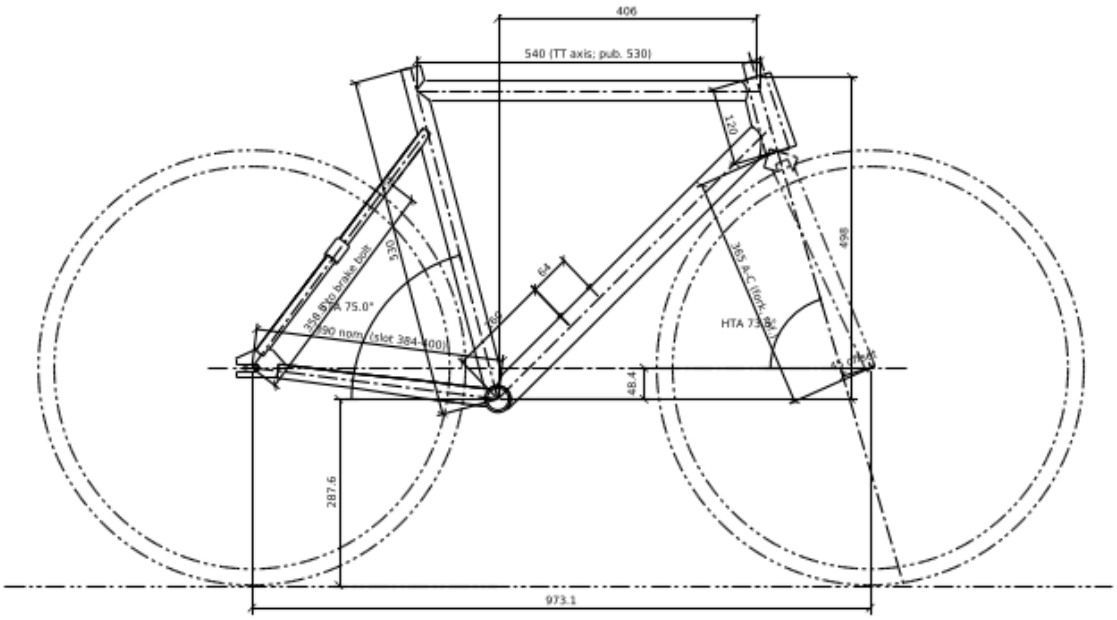
- Confirm hub (flip-flop or freewheel), crank and Gates sprocket part numbers so the chainline is exact.

- Pick the splitter style the builder prefers (seat-stay coupler vs. split dropout).

- Optional for a commuter: mudguard eyelets, rack mounts, dynamo routing - not included yet.

7. Published Hektor geometry, all sizes (reference)

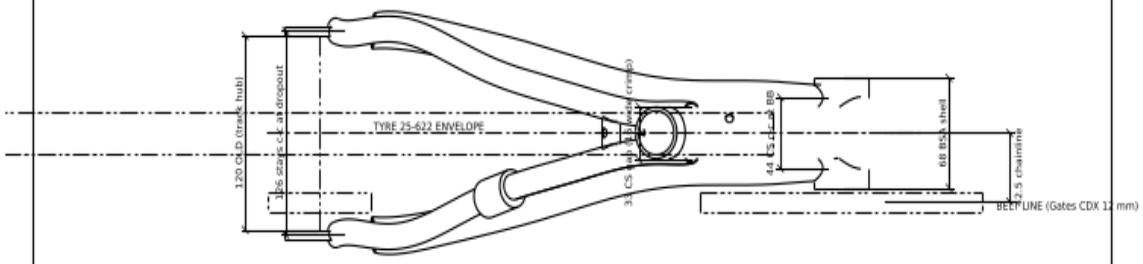
Size	Seat tube c-t (A)	Top tube horiz. (B)	Head tube (C)	Chainstay (D)	Head angle (E)	Seat angle (F)	Wheelbase (G)	Standover (H)	Reach (I)	Stack (J)
53 S	530	530	120	378	73.5	75.0	961	793	406	498
56 M	560	550	140	378	73.0	74.5	982	816	416	516
59 L	590	570	160	378	74.0	74.0	988	841	426	537
62 XL	620	590	180	378	74.0	74.0	1008	865	440	557



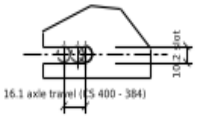
NOTES
1. Dimensions to tube centre lines; frame sized for 700c (622) wheels, 25 mm tyres.
2. Fit geometry = published Hektar 53 (stack 498, reach 406, HTA 73.5, STA 75.0). BB drop 48.4 derived for a 365/45 fork.
3. Horizontal TT measures 539 at stack height; published 530 is inconsistent with published stack/reach/STA (same 9-10 mm in all sizes).
4. Rear axle slot: chainstay 384-400. 60T/22T + 113T belt sits at 390.3. WB 961 at published CS 378 (not reachable with round seat tube).
5. Trail 52.6 mm. Toe overlap approx. 18 mm (170 cranks) - normal for track geometry.
6. Fork, wheels and sprockets shown as reference only (phantom). DXF is 1:1 - do not scale the PDF.

TITANIUM TRACK COMMUTER - SIZE 53 S
SIDE ELEVATION - FRAME GEOMETRY
Ti 3Al-2.5V / 6Al-4V PROJECTION: 1st angle
SHEET 1/2 SCALE 1:5 ON A3 UNITS mm / deg REV A

PLAN VIEW FROM ABOVE (front to the right, drive side down)



DROPOUT DETAIL (DRIVE SIDE, OUTSIDE VIEW)
6Al-4V plate, 6 mm, horizontal rear-facing slot
Axle marks: CS 384 / 390 (stock belt) / 400



NOTES
1. Chainstays 22.2 x 0.9 crimped to 16 x 27 oval from BB to 110 mm behind BB (clears 25 mm tyre AND 60T belt ring at 42.5 chainline).
2. Stays run at 63 mm from centre plane for the first 85 mm above/ahead of the dropout so the belt clears the drive-side stays.
3. Belt drive requires a frame split: bolted splitter in the drive-side seat stay (shown), or a split dropout per builder's standard.
4. Rear brake bridge: 12.7 x 1.0 tube, 6 mm bolt hole; reach 47-57 mm over slot (long-reach caliper).

TITANIUM TRACK COMMUTER - SIZE 53 S
REAR END PLAN VIEW + DROPOUT DETAIL
Ti 3Al-2.5V / 6Al-4V PROJECTION: 1st angle
SHEET 2/2 SCALE 1:2 ON A3 UNITS mm / deg REV A